

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
FALL 2025**



**THE KIOSK GUYS
SMART KIOSK**

**MIGUEL CERVANTES
PRAKASH SAPKOTA
EARL SANCHEZ
THIEN NGUYEN
IVAN KO**

REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	09.29.2025	MC	Document creation
1.0	10.17.2025	MC, ES	Document initialized
2.0	03.25.2026	MC, ES	Document revised
3.0	04.17.2026	MC	Document finalized

CONTENTS

1	Product Concept	8
1.1	Purpose and Use	8
1.2	Intended Audience	8
2	Product Description	9
2.1	Features & Functions	9
2.2	External Inputs & Outputs	10
2.3	Product Interfaces	11
3	Customer Requirements	15
3.1	Conference Room Scheduling System	15
3.1.1	Description	15
3.1.2	Source	15
3.1.3	Constraints	15
3.1.4	Standards	15
3.1.5	Priority	15
3.2	One-Tap Request Feature	15
3.2.1	Description	15
3.2.2	Source	16
3.2.3	Constraints	16
3.2.4	Standards	16
3.2.5	Priority	16
3.3	Search Functionality	16
3.3.1	Description	16
3.3.2	Source	16
3.3.3	Constraints	16
3.3.4	Standards	16
3.3.5	Priority	16
3.4	Smart Kiosk Display	16
3.4.1	Description	16
3.4.2	Source	16
3.4.3	Constraints	17
3.4.4	Standards	17
3.4.5	Priority	17
3.5	Remote Website Accessibility	17
3.5.1	Description	17
3.5.2	Source	17
3.5.3	Constraints	17
3.5.4	Standards	17
3.5.5	Priority	17
3.6	Idle Screen Display	17
3.6.1	Description	17
3.6.2	Source	17
3.6.3	Constraints	17
3.6.4	Standards	18
3.6.5	Priority	18

3.7	Swipe Card Functionality	18
3.7.1	Description	18
3.7.2	Source	18
3.7.3	Constraints	18
3.7.4	Standards	18
3.7.5	Priority	18
4	Packaging Requirements	19
4.1	Smart Kiosk Hardware delivery	19
4.1.1	Description	19
4.1.2	Source	19
4.1.3	Constraints	19
4.1.4	Standards	19
4.1.5	Priority	19
4.2	Software delivery	19
4.2.1	Description	19
4.2.2	Source	19
4.2.3	Constraints	19
4.2.4	Standards	20
4.2.5	Priority	20
5	Performance Requirements	21
5.1	Touchscreen Responsiveness	21
5.1.1	Description	21
5.1.2	Source	21
5.1.3	Constraints	21
5.1.4	Standards	21
5.1.5	Priority	21
5.2	Website stability	21
5.2.1	Description	21
5.2.2	Source	21
5.2.3	Constraints	21
5.2.4	Standards	21
5.2.5	Priority	21
5.3	User Login Speeds	21
5.3.1	Description	21
5.3.2	Source	22
5.3.3	Constraints	22
5.3.4	Standards	22
5.3.5	Priority	22
5.4	Reservation Confirmation Speeds	22
5.4.1	Description	22
5.4.2	Source	22
5.4.3	Constraints	22
5.4.4	Standards	22
5.4.5	Priority	22
5.5	Supply Request Automated Email	22
5.5.1	Description	22

5.5.2	Source	22
5.5.3	Constraints	22
5.5.4	Standards	22
5.5.5	Priority	22
5.6	Direction Mapping Speeds	23
5.6.1	Description	23
5.6.2	Source	23
5.6.3	Constraints	23
5.6.4	Standards	23
5.6.5	Priority	23
5.7	Screen Idling	23
5.7.1	Description	23
5.7.2	Source	23
5.7.3	Constraints	23
5.7.4	Standards	23
5.7.5	Priority	23
6	Safety Requirements	24
6.1	Laboratory equipment lockout/tagout (LOTO) procedures	24
6.1.1	Description	24
6.1.2	Source	24
6.1.3	Constraints	24
6.1.4	Standards	24
6.1.5	Priority	24
6.2	National Electric Code (NEC) wiring compliance	24
6.2.1	Description	24
6.2.2	Source	24
6.2.3	Constraints	24
6.2.4	Standards	24
6.2.5	Priority	24
7	Maintenance & Support Requirements	25
7.1	System Maintenance and Support	25
7.1.1	Description	25
7.1.2	Source	25
7.1.3	Constraints	25
7.1.4	Standards	25
7.1.5	Priority	25
8	Other Requirements	26
8.1	System Modularity and Extensibility	26
8.1.1	Description	26
8.1.2	Source	26
8.1.3	Constraints	26
8.1.4	Standards	26
8.1.5	Priority	26
8.2	Cross-Platform and Browser Compatibility	26
8.2.1	Description	26

8.2.2	Source	26
8.2.3	Constraints	26
8.2.4	Standards	27
8.2.5	Priority	27
9	Future Items	28
9.1	Mini-PC	28
9.1.1	Description	28
9.1.2	Source	28
9.1.3	Constraints	28
9.1.4	Standards	28
9.1.5	Priority	28

LIST OF FIGURES

1	Smart Kiosk conceptual drawing	8
2	Smart Kiosk lobby integration	9
3	ERSA Login Page	11
4	ERSA Dashboard	12
5	ERSA Dashboard #2	12
6	Admin Dashboard	13
7	ERB Interface	13
8	Remote Website	14

1 PRODUCT CONCEPT

This section describes the purpose, use and intended user audience for the Smart Kiosk. The Smart Kiosk is an interactive touchscreen device designed to enhance user convenience and accessibility for staff and students performing everyday tasks within the Engineering Research South A building (ERSA) and for centralized information and visitor navigation in the Engineering Research Building (ERB). The Smart Kiosk will improve the user experience by allowing users to reserve conference rooms, request supplies, and navigate the premises.

1.1 PURPOSE AND USE

The purpose of the Smart Kiosk is to provide a convenient and structured way for ERSA staff to manage room reservations, office supply requests, and room directory services. For the ERB, it will provide a centralized information board and navigation for students and visitors. The Smart Kiosk will be implemented through a touchscreen stand located in the building's lobby towards the entrance, allowing users to easily access it upon arrival. Furthermore, our solution aims to promote the newly established building and the CSE department by celebrating and advertising their social media accounts, and other CSE news.

1.2 INTENDED AUDIENCE

The Smart Kiosk is intended for UTA's Computer Science and Engineering (CSE) staff and students who have access to the Engineering Research South A building (ERSA) and to those who are visiting the Engineering Research Building (ERB). The Smart Kiosk and its accompanying websites are not designed for public or commercial use. The Smart Kiosk is intended for general use, but a larger emphasis has been given to accessibility, with older staff members specifically.

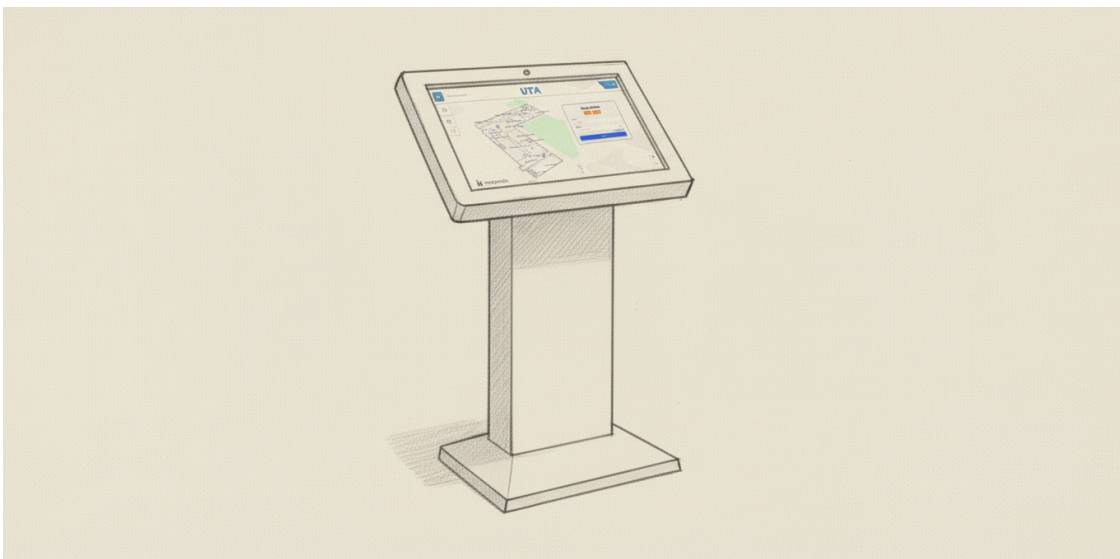


Figure 1: Smart Kiosk conceptual drawing



Figure 2: Smart Kiosk lobby integration

2 PRODUCT DESCRIPTION

This section provides the reader with an overview of the Smart Kiosk. The primary operational aspects of the Smart Kiosk, from the perspective of end users, maintainers, and administrators, are defined here. The key features and functions found in the Smart Kiosk, as well as critical user interactions and user interfaces, are described in detail in this section.

2.1 FEATURES & FUNCTIONS

The Smart Kiosk consists of both hardware and software components that are designed to function as ERSA's utility panel and the ERB's visitor navigation and information board. The device will feature a touchscreen display mounted on a tablet floor stand, running a web interface that streamlines common, mundane tasks for CSE staff and their visitors within the facility. The Smart Kiosk's web application will provide an accessible and visually engaging interface that integrates a conference room scheduling system, a one-tap request feature, and a search mapping function.

The Smart Kiosk system features a centralized announcement board that is shared across all kiosks. Admins will be able to activate or deactivate the banners for the kiosk via the kiosk itself or remotely. Adding, Editing or Deleting new items requires admins to navigate to the remote website.

The conference room scheduling functionality allows as users to create and delete room reservations directly from the smart kiosk or remotely from the remote website. Admins are able to delete room reservations for others via the provided admin dashboard either through the kiosk or remotely. After

successfully reserving a room, users will receive a confirmation email that allows them to add the reservation to their Google Calendar, Outlook Calendar, and Apple Calendar. Canceling a reservation will also send an email to the user providing relevant information such as the reservation that was canceled and whether it was canceled by the user or an admin.

The second main functionality of the kiosk's website is its ability to easily request supplies. Users will simply select supplies from the offered selections, and once they submit the request, the system will trigger an automated email to the ERSA's supply manager who will receive the list of supplies requested and by who it was requested.

The final core functionality would be the search mapping function, which provides a visual representation of the ERSA building's layout. This feature shows users how to get to their desired location in the building through a highlighted route.

Before users can access these core functionalities, they must be logged into their user account. Upon startup, the kiosk shall display a welcome page where the user is asked to either sign in or create an account. When making a profile, the user must input a valid UTA email. Placing restrictions on the email domain is one of the necessary authentication methods implemented to ensure this product is only for UTA staff and students. Another important feature that shall be implemented within the login page would be a way to quickly sign in to their profile. Our team recognized how inconvenient it may be for users to type in their entire email when accessing the kiosk repeatedly; therefore, users will be allowed to link their UTA ID card to their account, allowing quick and easy access to their account.

Many other features will be included in the website application however, the ones mentioned above are the most critical to achieving our sponsors' goals. On the contrary, some of the features that the Smart Kiosk will not be able to support include direct user-to-user communication, biometric authentication, progress tracking tasks, and access to any other applications/websites outside its own primary web application. These exclusions ensure a secure and focused user experience with our sponsor's intended purpose.

2.2 EXTERNAL INPUTS & OUTPUTS

The following table represents all of data inputs and outputs for the Kiosk.

Name	Description	Use
Sign Up	Input: User inputs email, name, unique username, hidden pin, and password	Add a new user to the database and gives the user access to the home page
Log In	Input: username, password, pin, card swipe	Quick log in for existing user to give them access to the home page
Interactive Map	Input: Location selection for routing, pin drop for routing, and inputs for interacting with the 3D map	Interaction with the map and routing system
Supply Notifier	Input: Selection from a list of items or input of a new item that is low on stock	User is able to request for supplies that are low on stock
Room Reservation	Input: Select room to reserve and date	User is able to reserve a conference room at selected times
Announcement Updater	Input: Add new announcements	Admin adds new announcements to announcements board
Email Notification	Output: Email sent to supply Restocker	When an item is requested, an email notification will be sent to the restocker
Routing to Rooms	Output: A highlighted path on the interactive map to the desired room	When a user requests direction to a room, the route will be provided on the map
User Login Response	Output: Email response to user's linked email	In response to successful sign in or forgotten passwords

2.3 PRODUCT INTERFACES

Below are some interfaces of the Smart Kiosk System.

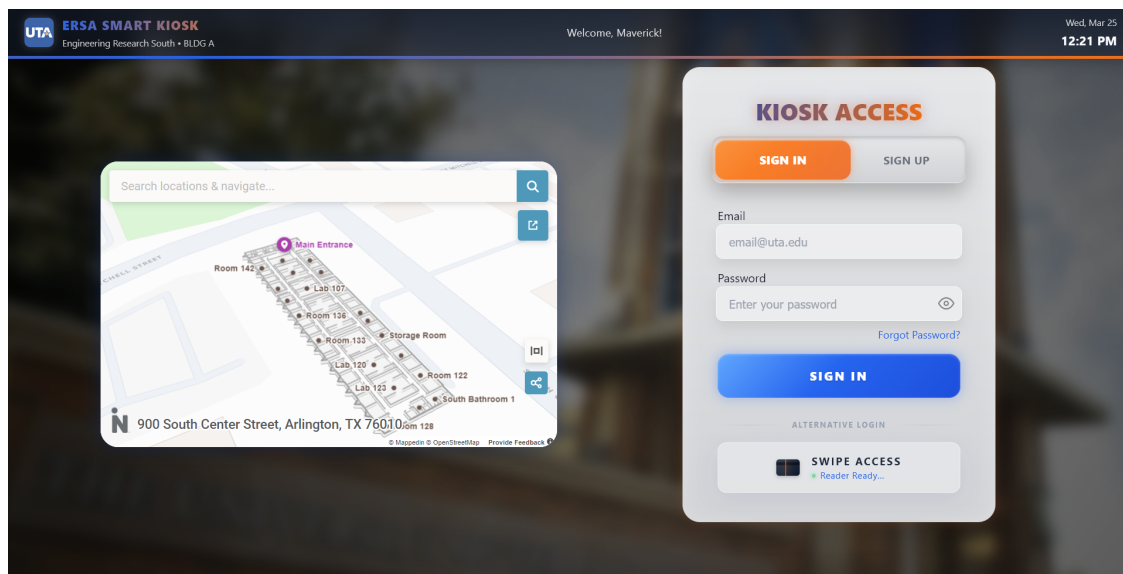


Figure 3: ERSa Login Page

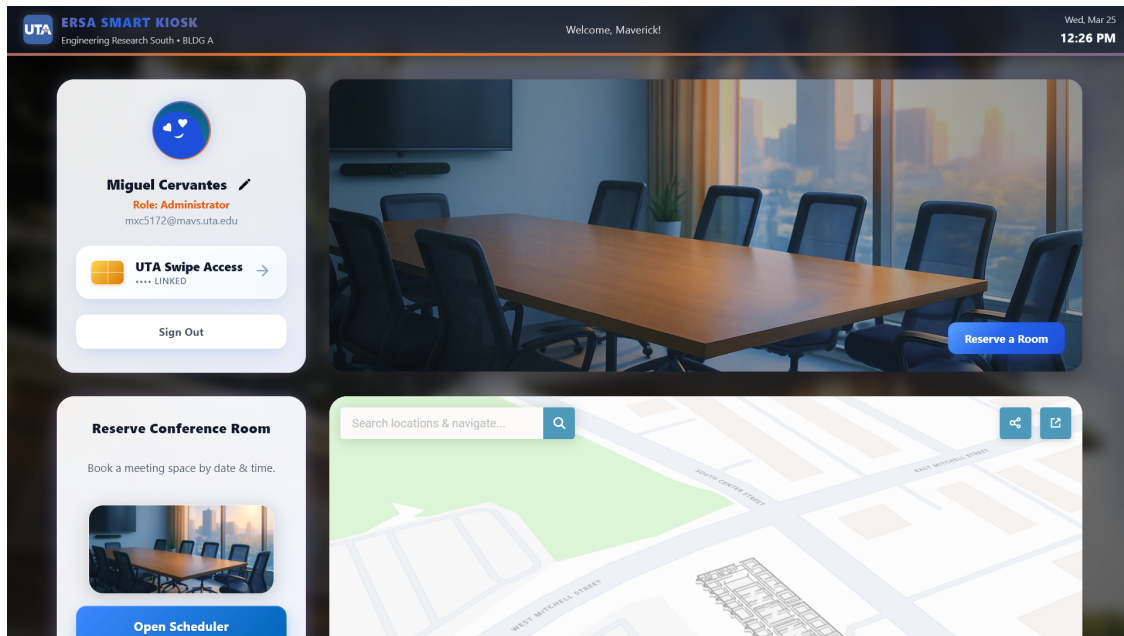


Figure 4: ERS Dashboard

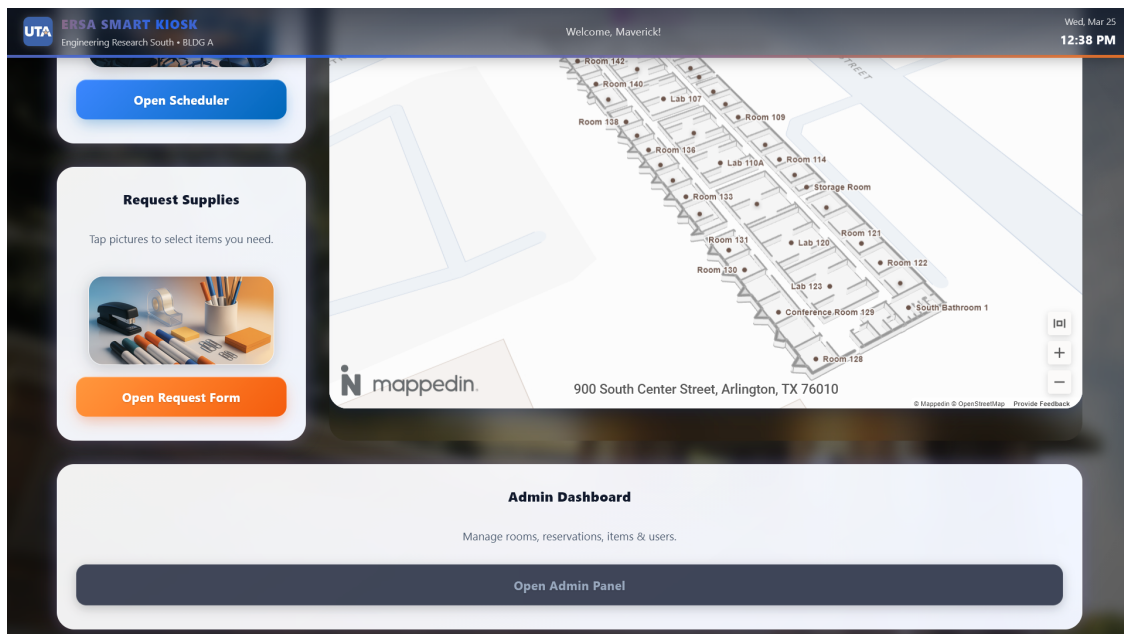


Figure 5: ERS Dashboard #2

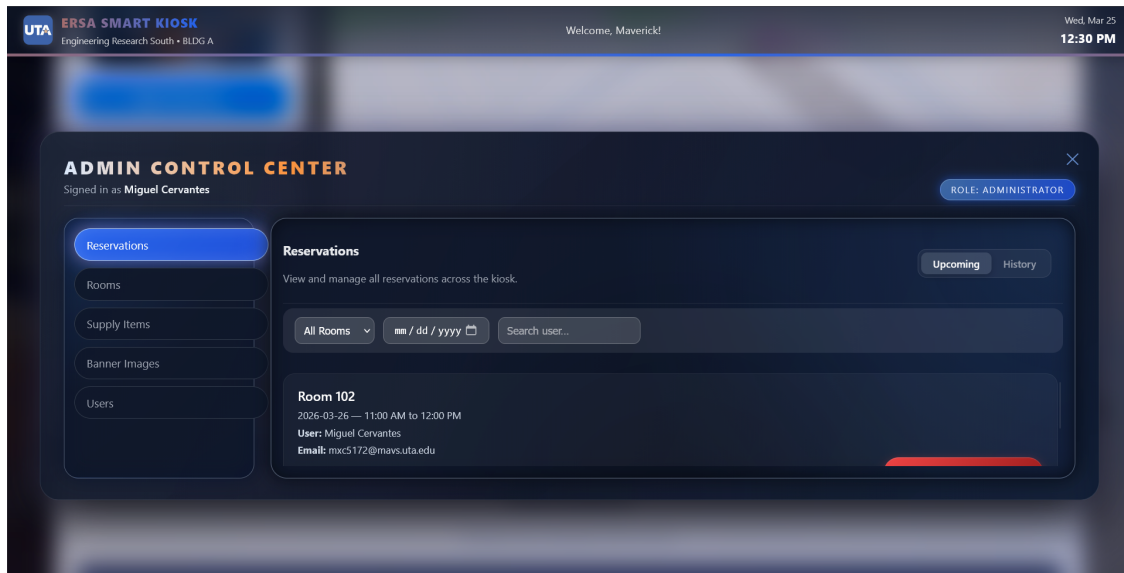


Figure 6: Admin Dashboard

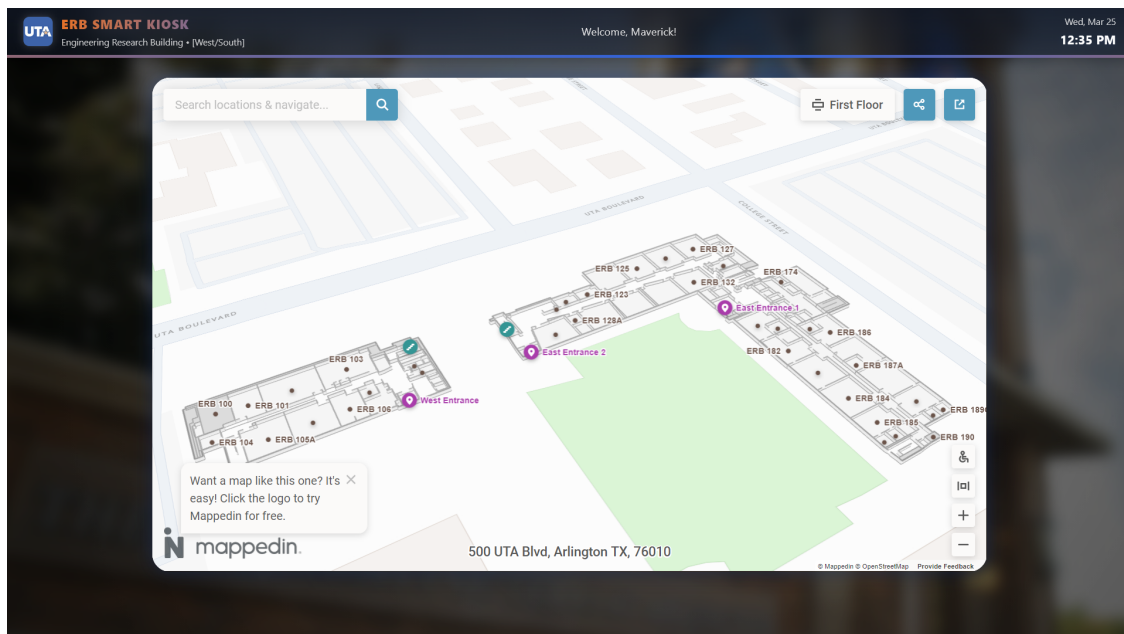


Figure 7: ERB Interface

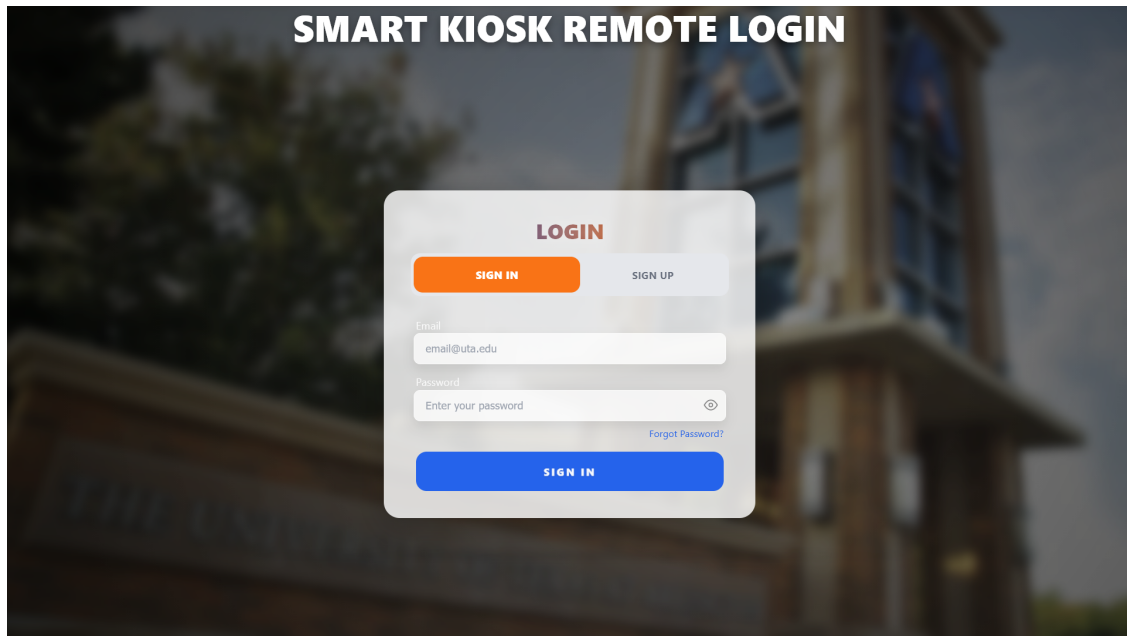


Figure 8: Remote Website

3 CUSTOMER REQUIREMENTS

The Smart Kiosk is an interactive touchscreen device that provides a convenient and efficient way to streamline everyday common tasks at the ERSA building and provide navigation and centralized information to visitors and students in the ERB building. This section established what each potential end-user should expect the product do and/or not do. The following customer requirements are essential features and functionalities of the product that must be met in order to satisfy the intended audience's needs.

3.1 CONFERENCE ROOM SCHEDULING SYSTEM

3.1.1 DESCRIPTION

The first major functionality of the Smart Kiosk shall allow end users to reserve the Conference Rooms located in the ERSA building. Users shall be able to select one of the two conference rooms they'd like to reserve and would be prompted to select from the available date(s) and time(s). Reservations shall be granted on a first-come-first-serve basis. A confirmation screen would appear to inform the user whether their reservation attempt was successful, along with email confirmation. This feature shall work in tandem with Outlook Calendar to allow users to easily see availability and update upon a successful reservation.

3.1.2 SOURCE

This requirement comes from the product owner and project sponsor, Jade Myers. Myers is the CSE communications assistant at UTA.

3.1.3 CONSTRAINTS

This functionality won't allow conference room reservations to overlap in regards to time. The system shall also not allow standard users to modify existing reservations that are not their own. Old reservation records shall be stored for a month before being deleted from the database. Users cannot utilize this feature unless signed in to their profile.

3.1.4 STANDARDS

ISO 9241-210

ISO 8601

3.1.5 PRIORITY

Critical Priority

3.2 ONE-TAP REQUEST FEATURE

3.2.1 DESCRIPTION

This functionality of the Smart Kiosk shall provide users a simple way to notify administrator users whenever office, break room, and bathroom supplies are needed. Users shall be given a selection of supplies in the form of large image icons that they can choose from. Tapping the icon would indicate a selection/deselection of the supply. After selecting all the necessary supplies, the user would submit the request, which would trigger the kiosk to automatically email the supply manager. This feature shall also track the frequency of user requests to accurately predict when supplies are running low, resulting in proactive notifications

3.2.2 SOURCE

This requirement comes from the project sponsors, Jade Myers and Joseph 'JD' Briggs. Myers is the product owner, while Briggs is one of the supply managers in charge at the ERSA building.

3.2.3 CONSTRAINTS

This functionality will not specify the amount of supplies that should be delivered to the ERSA building. Additionally, the specific location to which these supplies will be delivered to will not be mentioned.

3.2.4 STANDARDS

ISO 9241-210:2019

WCAG 2.1

ADA Title III

3.2.5 PRIORITY

Critical Priority

3.3 SEARCH FUNCTIONALITY

3.3.1 DESCRIPTION

The Smart Kiosk shall provide users with the ability to obtain directions to their desired destination upon selection. Users shall be able to select a room source and destination from a map of ERSA or ERB through search or selection functionalities. Upon selecting the rooms they'd like to create a route for, a highlighted path shall be displayed showcasing a direct route they can navigate to.

3.3.2 SOURCE

This requirement comes from the product owner and project sponsor, Jade Myers. Myers is the CSE communications assistant at UTA.

3.3.3 CONSTRAINTS

The mapping feature shall not provide step-by-step written directions or numeric distance values.

3.3.4 STANDARDS

ISO 9241-110:2020

ISO 9241-171:2008

WCAG 2.1

ADA Title III Compliance

3.3.5 PRIORITY

Critical

3.4 SMART KIOSK DISPLAY

3.4.1 DESCRIPTION

The Smart Kiosk shall feature an LCD touchscreen display that is large enough to allow users to easily view and interact with on-screen content.

3.4.2 SOURCE

This requirement comes from the product owner and project sponsor, Jade Myers. Myers is the CSE communications assistant at UTA.

3.4.3 CONSTRAINTS

The screen must be touch compatible. The screen size must be larger than 13.6 inches. The display must feature a 1080P resolution or better.

3.4.4 STANDARDS

ISO 9241-303:2011

ISO 13406-2

IEC 60950-1 / IEC 62368-1

3.4.5 PRIORITY

Critical

3.5 REMOTE WEBSITE ACCESSIBILITY

3.5.1 DESCRIPTION

In addition to the Smart Kiosk, ERSA users shall be able to access a limited version of the website remotely. These users will be able to login to the website using their same account credentials as the kiosk, and have the ability to reserve conference rooms from their own devices. If they are an admin, they will have access to the whole distributed kiosk system.

3.5.2 SOURCE

This requirement comes from the product owner and project sponsor, Jade Myers. Myers is the CSE communications assistant at UTA.

3.5.3 CONSTRAINTS

The remote website must require users to be connected to the internet. Using the website requires an account. Furthermore, the remote website is hosted on Render on a paid plan, which requires a recurring monthly payment to maintain availability.

3.5.4 STANDARDS

WCAG 2.1 ISO 9241-210:2019 ISO/IEC 23026:2015 ISO/IEC 25010:2011

3.5.5 PRIORITY

Critical

3.6 IDLE SCREEN DISPLAY

3.6.1 DESCRIPTION

While the kiosk is currently not in use, the screen shall display an idle screensaver featuring UTA CSE-related content, including logos, pictures, social media handles, and the latest announcements. The idle screen shall automatically activate after 1 minute of inactivity and shall be interrupted whenever the Smart Kiosk detects user input.

3.6.2 SOURCE

This requirement comes from the product owner and project sponsor, Jade Myers. Myers is the CSE communications assistant at UTA.

3.6.3 CONSTRAINTS

No constraints available at the moment.

3.6.4 STANDARDS

UTA Intellectual Property Policy, HOP Policy 5-702

3.6.5 PRIORITY

Moderate

3.7 SWIPE CARD FUNCTIONALITY

3.7.1 DESCRIPTION

The Smart Kiosk shall provide a secure way for users to sign in to their profile by swiping their UTA ID card into the card reader. This provides a convenient way to access the kiosk.

3.7.2 SOURCE

This requirement comes from the product owner and project sponsor, Jade Myers. Myers is the CSE communications assistant at UTA.

3.7.3 CONSTRAINTS

Users must have their UTA ID card to utilize this login feature.

3.7.4 STANDARDS

ANSI/INCITS 322-2008

NIST SP 800-63B

ISO/IEC 27001:2022

ISO/IEC 29115:2013

ISO 9241-210:2019

3.7.5 PRIORITY

Low

4 PACKAGING REQUIREMENTS

This section outlines the packaging requirements for how the Smart Kiosk and its website will be delivered and installed to the end-users. These requirements are valuable as they ensure that the product is ready for immediate use upon deployment. There are two main components that will be delivered. The freestanding kiosk that the users will physically interact with, and the website that the user will be interacting with through the Smart Kiosk. These requirements are valuable as they ensure that the product is ready for immediate use upon deployment.

4.1 SMART KIOSK HARDWARE DELIVERY

4.1.1 DESCRIPTION

The Smart Kiosk will be fully assembled and tested before delivery to the ERSA and ERB buildings. Once the product has been built and its functionality has been verified by the team, it shall be packaged using a pallet and carton method. This method provides a firm layer of protection and stability during transportation at a reasonable price point. A pallet jack will be used to transport the hardware to and from a van, which is the selected form of transportation. Once delivered and stationed at the desired location, our team would simply just plug in the power supply for the kiosk and connect the system with the correct WiFi connectivity.

4.1.2 SOURCE

The Kiosk Guys

4.1.3 CONSTRAINTS

The Smart Kiosk must be completely packaged and secure before transportation. Use of the Smart Kiosk must remain indoors at the ERSA and ERB building. No food and drinks shall be near the Smart Kiosk upon use.

4.1.4 STANDARDS

List of applicable standards

4.1.5 PRIORITY

Critical

4.2 SOFTWARE DELIVERY

4.2.1 DESCRIPTION

The Smart Kiosk will access a website for all its functionality. The necessary software files and settings will already be retrieved from the GitHub repository and configured on the Raspberry Pi before delivery. There is no need for the user to set up anything, as the machine would be configured to automatically boot up the application on the computer's local server. A link to the GitHub repository, instructions, and user manual would be provided to the system administrators along with our product owners and project sponsors. Upon signing up as a new user, a user manual to explain all Kiosk functionalities will be provided in the sign up confirmation email sent to the user.

4.2.2 SOURCE

The Kiosk Guys

4.2.3 CONSTRAINTS

Software used for the Smart Kiosk must be compatible with the internal computer. The local web server shall operate on ports 8000 and 5173. An internet connection must be established for use. Only

administrative users shall have permission to modify system settings. The end-users shall not have access to any other computer applications or websites not related to the product.

4.2.4 STANDARDS

Sensitive information must be transferred securely. Licensing terms of the software used must be properly documented.

4.2.5 PRIORITY

Critical

5 PERFORMANCE REQUIREMENTS

This section provides an in-depth look at the performance requirements of the Smart Kiosk. Performance requirements focus on the reliability of the Smart Kiosk and the speeds at which specific critical operations must operate.

5.1 TOUCHSCREEN RESPONSIVENESS

5.1.1 DESCRIPTION

The touchscreen will need to be highly responsive and sensitive in order to operate effectively. The touch latency of the display should be under 100 ms to ensure users have a smooth and instantaneous experience. The touchscreen display will include multi-touch support in order to detect different types of contact and gestures being performed by the user.

5.1.2 SOURCE

The Kiosk Guys

5.1.3 CONSTRAINTS

Touch screen latency may vary depending on how CPU intensive the task may be.

5.1.4 STANDARDS

N/A

5.1.5 PRIORITY

Critical

5.2 WEBSITE STABILITY

5.2.1 DESCRIPTION

Website stability and reliability is crucial when determining how successful the Smart Kiosk will be. Without a reliable website, the Smart Kiosk will not be able to operate. The system must maintain an uptime rate of at least 99 percent to ensure the end-users are able to access it when they desire without any disruptions.

5.2.2 SOURCE

The Kiosk Guys

5.2.3 CONSTRAINTS

Connection to the website for the Smart Kiosk shall be dependent on its WiFi connection to ERSA's and ERB's building internet router.

5.2.4 STANDARDS

N/A

5.2.5 PRIORITY

Critical

5.3 USER LOGIN SPEEDS

5.3.1 DESCRIPTION

The system must be able to successfully authenticate and verify the details entered by the user at login with the database within 3 seconds of the user's attempt.

5.3.2 SOURCE

The Kiosk Guys

5.3.3 CONSTRAINTS

The website application must approve the log in attempt if the user provided a matching username (email, pin, etc) and password, and deny if otherwise.

5.3.4 STANDARDS

N/A

5.3.5 PRIORITY

Critical

5.4 RESERVATION CONFIRMATION SPEEDS

5.4.1 DESCRIPTION

The Smart Kiosk must be able to confirm whether or not a reservation for a conference room has been successfully placed or edited within 1 second of a user request.

5.4.2 SOURCE

The Kiosk Guys

5.4.3 CONSTRAINTS

The reservation confirmation speed must be met regardless of the user's operation (reservation creation, deletion and alteration)

5.4.4 STANDARDS

SMTP

5.4.5 PRIORITY

Critical

5.5 SUPPLY REQUEST AUTOMATED EMAIL

5.5.1 DESCRIPTION

Upon a successful supply request by a user, the Smart Kiosk shall send an automated email to the administrative users almost instantaneously. The general format of the email must always be the same, only changing based on the items requested by the user.

5.5.2 SOURCE

The Kiosk Guys

5.5.3 CONSTRAINTS

N/A

5.5.4 STANDARDS

SMTP

5.5.5 PRIORITY

Critical

5.6 DIRECTION MAPPING SPEEDS

5.6.1 DESCRIPTION

The Smart Kiosk shall display a highlighted route to the user after a valid mapping request under 1 second.

5.6.2 SOURCE

The Kiosk Guys

5.6.3 CONSTRAINTS

The Smart Kiosk website utilizes MappedIn for the mapping functionalities, therefore its functionality will depend on them.

5.6.4 STANDARDS

N/A

5.6.5 PRIORITY

Critical

5.7 SCREEN IDLING

5.7.1 DESCRIPTION

The Smart Kiosk enters an idle state where it displays a set of pre-selected screens after 1 minute of total inactivity. After 30 seconds of inactivity, a prompt is displayed to the user saying that the screen will go idle and, their profile will be logged out. This prompt will start a countdown of 30 seconds and after that countdown the user will be logged out.

5.7.2 SOURCE

The Kiosk Guys

5.7.3 CONSTRAINTS

The screens used for the Smart Kiosk's idle states will require images to be updated frequently by the kiosk administrators

5.7.4 STANDARDS

N/A

5.7.5 PRIORITY

Moderate

6 SAFETY REQUIREMENTS

This section provides a detailed look at the safety requirements that must be in place during the development and post-development of the Smart Kiosk to ensure the safety of the system, its environment, and its users.

6.1 LABORATORY EQUIPMENT LOCKOUT/TAGOUT (LOTO) PROCEDURES

6.1.1 DESCRIPTION

Any fabrication equipment provided used in the development of the project shall be used in accordance with OSHA standard LOTO procedures. Locks and tags are installed on all equipment items that present use hazards, and ONLY the course instructor or designated teaching assistants may remove a lock. All locks will be immediately replaced once the equipment is no longer in use.

6.1.2 SOURCE

CSE Senior Design laboratory policy

6.1.3 CONSTRAINTS

Equipment usage, due to lock removal policies, will be limited to availability of the course instructor and designed teaching assistants.

6.1.4 STANDARDS

Occupational Safety and Health Standards 1910.147 - The control of hazardous energy (lockout/tagout).

6.1.5 PRIORITY

Critical

6.2 NATIONAL ELECTRIC CODE (NEC) WIRING COMPLIANCE

6.2.1 DESCRIPTION

Any electrical wiring must be completed in compliance with all requirements specified in the National Electric Code. This includes wire runs, insulation, grounding, enclosures, over-current protection, and all other specifications.

6.2.2 SOURCE

CSE Senior Design laboratory policy

6.2.3 CONSTRAINTS

High voltage power sources, as defined in NFPA 70, will be avoided as much as possible in order to minimize potential hazards.

6.2.4 STANDARDS

NFPA 70

6.2.5 PRIORITY

Critical

7 MAINTENANCE & SUPPORT REQUIREMENTS

The Smart Kiosk system requires ongoing maintenance and support to ensure reliability, usability, and security after deployment. Maintenance activities shall address both software-related issues (e.g., the Smart Kiosk website and backend services) and hardware-related issues (e.g., display, touchscreen, and card reader). The project team shall provide appropriate documentation, access permissions, and troubleshooting procedures to maintain the product effectively once deployed in the field.

7.1 SYSTEM MAINTENANCE AND SUPPORT

7.1.1 DESCRIPTION

The Smart Kiosk system shall be maintained by the development team after deployment. The team will retain full access to the GitHub repository that hosts the website and backend codebase, allowing for remote troubleshooting, updates, and bug fixes. In the event of software-related issues, authorized maintainers shall remotely push patches or configuration updates through the existing version control system. For hardware-related issues (e.g., display malfunction, card reader failure, or connectivity problems), maintenance personnel shall refer to the provided Instruction and Troubleshooting Manual. If the issue cannot be resolved through documentation, the development team shall perform an on-site inspection and repair as necessary.

7.1.2 SOURCE

The Kiosk Guys
Jade Myers

7.1.3 CONSTRAINTS

Maintenance activities will be limited to authorized personnel with access credentials to the GitHub repository and administrative privileges on the kiosk system. Physical maintenance shall depend on the availability of replacement parts and access to the kiosk location. Remote software maintenance will require reliable internet access to the kiosk network.

7.1.4 STANDARDS

ISO/IEC 14764:2006
ISO/IEC 12207:2017
IEEE 828-2012
ISO/IEC 27001:2022
IEEE 16326-2009

7.1.5 PRIORITY

High

8 OTHER REQUIREMENTS

The Smart Kiosk system shall be designed with modularity, portability, and extensibility in mind to ensure long-term maintainability and scalability. This section defines the requirements that support future enhancements, system reconfiguration, and cross-platform compatibility. These requirements ensure that the Smart Kiosk and its associated web application can be easily updated, deployed, and integrated with new technologies or environments without major redesign.

8.1 SYSTEM MODULARITY AND EXTENSIBILITY

8.1.1 DESCRIPTION

The Smart Kiosk system architecture shall be modular, allowing individual components (e.g., map navigation, one-tap requests, swipe card login) to be updated or replaced independently without affecting the overall functionality of the system. The software design shall support future extensibility, enabling the addition of new features such as updated UI layouts, integration with other university systems, or additional data analytics modules without major structural changes to the existing codebase.

8.1.2 SOURCE

The Kiosk Guys

8.1.3 CONSTRAINTS

All modules shall adhere to standardized interfaces and communication protocols to maintain system stability. Changes or additions must be reviewed and merged through the GitHub repository to preserve version integrity and prevent compatibility issues.

8.1.4 STANDARDS

ISO/IEC/IEEE 42010:2011

ISO/IEC 12207:2017

IEEE 1471-2000

ISO/IEC 25010:2011

8.1.5 PRIORITY

High

8.2 CROSS-PLATFORM AND BROWSER COMPATIBILITY

8.2.1 DESCRIPTION

The Smart Kiosk website shall be platform-independent and accessible from any major operating system, including Windows, macOS, and Linux. It shall also maintain full compatibility across modern browsers (e.g., Chrome, Firefox, Safari, and Edge). The system's source code shall be portable and deployable in both local and cloud-hosted environments without requiring modification.

8.2.2 SOURCE

This requirement originates from the development team's design goals and the sponsor's request for wide accessibility and maintainability.

8.2.3 CONSTRAINTS

Portability may be limited by third-party libraries or frameworks used in development. Web functionality shall depend on browser compliance with modern HTML5, CSS3, and JavaScript standards.

8.2.4 STANDARDS

W3C HTML5, CSS3, and ECMAScript Standards ISO/IEC 25010:2011 ISO/IEC 30170:2012 IEEE 1061-1998

8.2.5 PRIORITY

Moderate

9 FUTURE ITEMS

9.1 MINI-PC

9.1.1 DESCRIPTION

The internal PC of the Smart Kiosk will be swapped out to have a more powerful internal PC capable of handling the website smoothly.

9.1.2 SOURCE

The Kiosk Guys

9.1.3 CONSTRAINTS

Computing devices must be approved for purchase through OIT before ordering.

9.1.4 STANDARDS

IEEE 802.11

FCC Part 15

ACPI

RoHS (Directive 2011/65/EU)

9.1.5 PRIORITY

Low

REFERENCES